

Bao, Guanhao, et al. "Revealing the Correlation between Economic Indicators and Gold Prices for Forecasting: Medium Term Forecast Framework with Data Patch." *Expert Systems with Applications*, vol. 298, 2026, p. 129594, doi:10.1016/j.eswa.2025.129594.

Summary:

Researchers developed a new AI model, DPformer, to predict gold prices 5-30 days into the future using economic indicators. The model breaks gold prices into long-term trends and short-term seasonal patterns, then uses six major economic factors to predict future prices.

Methods:

- Model: DPformer (Decomposition-Patch Transformer) - separates gold prices into trend component (driven by macroeconomics) and seasonal component (short-term fluctuations)
- Data: Daily gold prices from January 2003 to December 2023 (5,473 observations)
- Economic indicators used: 10-year Treasury yield, US Dollar Index, S&P 500, Federal Funds Rate, Economic Policy Uncertainty, and Inflation Expectations
- Testing: 70% training data, 10% validation, 20% testing; forecasts at 5, 10, 20, and 30-day horizons
- Comparison: Tested against 10 other models, including Transformer, LSTM, and PatchTST

Results:

- 30-day forecast accuracy: Mean error of only 2.19% (predicted prices within ~\$40 of actual)
- Beat PatchTST by 14%, Transformer by 86%, and LSTM by 62%
- 5-day forecast: Even better at 0.68% error vs competitors' 1-8% error
- Successfully tracked major events like the COVID-19 crash (March 2020) and the Russia-Ukraine spike (March 2022)
- During Russia-Ukraine war, predicted March 8 peak at \$2,071.70 vs actual \$2,052.65 (only \$19 off)

Key/Important Findings:

- Stock market and Treasury yields drive gold prices most: S&P 500 and 10-year Treasury rates account for over 75% of what moves gold prices
- When stocks go up, gold goes down: High stock prices reduce demand for gold as a safe haven

- When interest rates go up, gold goes down: Higher yields make bonds more attractive than gold
- The model works especially well during volatile periods when other models fail badly
- Breaking prices into trends and seasonal patterns is crucial - models that don't do this make much bigger errors

Azimli, Asil. "Is gold a safe haven for the U.S. dollar during extreme conditions?" *International Economics*, vol. 177, 2024, p. 100478, doi:10.1016/j.inteco.2024.100478.

Summary:

This study examines whether gold protects investors from currency risk when the dollar weakens or strengthens against 14 other currencies, particularly during the COVID-19 pandemic and the Russia-Ukraine war. The focus is on whether gold is a "hedge" (works during normal times) or "safe haven" (works during crises).

Methods:

- Currency pairs examined: 14 total - including Chinese yuan, Indian rupee, Brazilian real, Russian ruble, Mexican peso, South Korean won, Turkish lira, plus major developed currencies (euro, pound, yen, Swiss franc, etc.)
- Time period: January 2017 to August 2023
- Analysis tools:
 - DCC-GARCH (measures changing correlations over time)
 - Quantile regression (examines behavior during extreme movements)
 - Portfolio optimization (calculates ideal gold allocation)
- Crisis periods analyzed: the COVID-19 pandemic and the Russia-Ukraine war

Results:

According to the preview findings stated in the introduction (couldn't access full paper):

- Gold works as a hedge for all currency pairs except USD/Chinese yuan
- Stronger protection for emerging market currencies than developed currencies
- COVID-19 period: Gold was a strong safe-haven for the Brazilian real, Russian ruble, Mexican peso, Norwegian krone; a weak safe-haven for the South Korean won and Indian rupee
- Russia-Ukraine war (first 2 months): Safe-haven for Brazilian real, Russian ruble, Indian rupee, Swiss franc, euro, Norwegian krone, Australian dollar
- War effect weakens over time: Safe-haven properties decline after 6 months
- Adding gold to currency portfolios reduces risk, but hedging costs increased significantly during both crises

Key/Important Findings:

- Gold protects better against emerging market currency risk than developed market currency risk

- Crisis type matters: Different currencies need gold protection during different crises
- During the initial COVID-19 panic (March 2020), holding gold actually increased portfolio variance for most pairs (opposite of expected)
- Time horizon is critical: Gold's protective benefits can strengthen or weaken as crises evolve
- The cost of using gold as protection (hedge ratios) went up substantially during crisis periods
- Gold provides "considerable" protection for the USD against the Brazilian real, Russian ruble, Mexican peso, and Turkish lira

Tanin, Tauhidul Islam, et al. "Do currency exchange rates impact gold prices? New evidence from the ongoing COVID-19 period." *International Review of Financial Analysis*, vol. 77, 2021, p. 101868, doi:10.1016/j.irfa.2021.101868.

Summary:

Researchers investigated whether currency movements affected gold prices during COVID-19 and compared this to the pre-pandemic period. They examined seven major currencies to see whether their relationships with gold changed during the crisis.

Methods:

- Currencies examined: US dollar vs British pound, euro, Canadian dollar, Japanese yen, Swiss franc, Norwegian krone, Danish krone
- Time periods:
 - COVID period: January 2020 to April 2021 (67 weeks)
 - Normal period: January 2017 to December 2019 (156 weeks)
- Statistical method: NARDL (Nonlinear Autoregressive Distributed Lag) - captures asymmetric effects where currency increases and decreases have different impacts
- Data: Daily prices grouped into weekly observations to smooth out noise

Results:

- Long-term: No currencies showed lasting impact on gold prices (neither during COVID nor before)
- Short-term during COVID:
 - British pound, Canadian dollar, Japanese yen, and Danish krone are getting stronger → gold prices went up
 - Euro getting weaker → gold prices went up
 - But 1-2 weeks later, the euro is getting weaker → gold prices went down (delayed adjustment)
- Comparison to pre-COVID: The relationship was much more unpredictable and uneven during COVID compared to the stable pre-pandemic period
- Gold price behavior during COVID: Jumped from \$1,500 to over \$2,000, with a sharp drop in March-April 2020 during peak panic
- Market efficiency: Past gold prices don't predict future gold prices (supports random walk theory)

Key/Important Findings:

- COVID-19 disrupted normal currency-gold relationships: What worked before the pandemic didn't work the same way during it

- No long-term predictability = gold is a safe haven: The fact that currencies can't predict gold prices long-term confirms that gold maintains its value independently
- Financial markets need time to adjust: Delayed effects (1-2 weeks later) show markets don't react instantly to shocks
- Swiss franc and Norwegian krone had zero impact on gold during COVID → these currencies might be good diversification tools alongside gold
- The gold market is "efficient" - you can't use past prices to predict future prices, even during a major crisis
- The study confirms gold served as a safe-haven asset during COVID-19 despite short-term volatility